a Boson: strict-cone screening; group-only time-reversal fallback

columns: $J_2^T g J_2$, J_2^2, and J_1, J_2 commuting (comm.) or anticommuting (anti.)

rows: $J_1^T g J_1$, J_1^2

g, I — comm.	g, I — anti.	g, -I — comm.	g, -I — anti.	-g, I — comm.	-g, I — anti.	-g, -I — comm.	-g, -I — anti.
B01 $d_{alg}=4$ group only no theorem from original J_1, J_2	B02 $d_{alg}=4$ strict matrix cone $\eta = ij_2g$ fixed-N weight	B03 $d_{alg}=2$ group only no theorem from original J_1, J_2	B04 $d_{alg}=4$ group only $\Theta(+,-)$: $W_B = \lvert w\rvert^2$	B05 $d_{alg}=2$ strict cone $\eta = J_2g$ semigroup branch	B06 $d_{alg}=4$ group only no theorem from original J_1, J_2	B07 $d_{alg}=4$ group only $\Theta(+,-)$: $W_B = \lvert w\rvert^2$	B08 $d_{alg}=4$ group only no theorem from original J_1, J_2
B09 $d_{alg}=2$ group only no theorem from original J_1, J_2	B10 $d_{alg}=4$ strict matrix cone $\eta = ij_2g$ fixed-N weight	B11 $d_{alg}=4$ group only no theorem from original J_1, J_2	B12 $d_{alg}=4$ group only $\Theta(-,-)$: Kramers physical $d_N = 8$	B13 $d_{alg}=4$ strict cone $\eta = J_2g$ semigroup branch	B14 $d_{alg}=2$ group only no theorem from original J_1, J_2	B15 $d_{alg}=4$ group only $\Theta(-,-)$: Kramers physical $d_N = 4$	B16 $d_{alg}=4$ group only no theorem from original J_1, J_2
B17 $d_{alg}=2$ strict matrix cone $\eta = ij_2g$ fixed-N weight	B18 $d_{alg}=4$ group only no theorem from original J_1, J_2	B19 $d_{alg}=4$ group only $\Theta(+,-)$: $W_B = \lvert w\rvert^2$	B20 $d_{alg}=2$ group only no theorem from original J_1, J_2	B21 $d_{alg}=4$ group only no theorem from original J_1, J_2	B22 $d_{alg}=2$ strict cone $\eta = J_2g$ semigroup branch	B23 $d_{alg}=4$ group only no theorem from original J_1, J_2	B24 $d_{alg}=4$ group only $\Theta(+,-)$: $W_B = \lvert w\rvert^2$
B25 $d_{alg}=4$ strict matrix cone $\eta = ij_2g$ fixed-N weight	B26 $d_{alg}=4$ group only no theorem from original J_1, J_2	B27 $d_{alg}=4$ group only $\Theta(-,-)$: Kramers physical $d_N = 4$	B28 $d_{alg}=4$ group only no theorem from original J_1, J_2	B29 $d_{alg}=4$ group only no theorem from original J_1, J_2	B30 $d_{alg}=4$ strict cone $\eta = J_2g$ semigroup branch	B31 $d_{alg}=8$ group only no theorem from original J_1, J_2	B32 $d_{alg}=4$ group only $\Theta(-,-)$: Kramers physical $d_N = 8$

b Fermion: strict-cone screening; group-only time-reversal fallback

columns: $J_2^T g J_2$, J_2^2, and J_1, J_2 commuting (comm.) or anticommuting (anti.)

rows: $J_1^T g J_1$, J_1^2

g, I — comm.	g, I — anti.	g, -I — comm.	g, -I — anti.	-g, I — comm.	-g, I — anti.	-g, -I — comm.	-g, -I — anti.
F01 $d_{alg}=2$ group only no theorem from original J_1, J_2	F02 $d_{alg}=4$ group only no theorem from original J_1, J_2	F03 $d_{alg}=4$ group only no theorem from original J_1, J_2	F04 $d_{alg}=4$ strict cone $\eta = ij_2g$ semigroup branch	F05 $d_{alg}=2$ group only no theorem from original J_1, J_2	F06 $d_{alg}=2$ group only no theorem from original J_1, J_2	F07 $d_{alg}=2$ strict cone $\eta = J_2g$ semigroup branch	F08 $d_{alg}=2$ group only no theorem from original J_1, J_2
F09 $d_{alg}=4$ group only no theorem from original J_1, J_2	F10 $d_{alg}=4$ group only $\Theta(+,-)$: $W_F \geq 0$	F11 $d_{alg}=4$ group only no theorem from original J_1, J_2	F12 $d_{alg}=8$ strict cone $\eta = ij_2g$ semigroup branch	F13 $d_{alg}=4$ group only $\Theta(+,-)$: $W_F \geq 0$	F14 $d_{alg}=4$ group only no theorem from original J_1, J_2	F15 $d_{alg}=4$ strict cone $\eta = J_2g$ semigroup branch	F16 $d_{alg}=4$ group only no theorem from original J_1, J_2
F17 $d_{alg}=2$ group only no theorem from original J_1, J_2	F18 $d_{alg}=2$ group only no theorem from original J_1, J_2	F19 $d_{alg}=4$ strict cone $\eta = ij_2g$ semigroup branch	F20 $d_{alg}=4$ group only no theorem from original J_1, J_2	F21 $d_{alg}=4$ group only no theorem from original J_1, J_2	F22 $d_{alg}=4$ group only no theorem from original J_1, J_2	F23 $d_{alg}=4$ group only no theorem from original J_1, J_2	F24 $d_{alg}=2$ strict cone $\eta = J_2g$ semigroup branch
F25 $d_{alg}=2$ group only $\Theta(+,-)$: $W_F \geq 0$	F26 $d_{alg}=2$ group only no theorem from original J_1, J_2	F27 $d_{alg}=4$ strict cone $\eta = ij_2g$ semigroup branch	F28 $d_{alg}=4$ group only no theorem from original J_1, J_2	F29 $d_{alg}=4$ group only no theorem from original J_1, J_2	F30 $d_{alg}=2$ group only $\Theta(+,-)$: $W_F \geq 0$	F31 $d_{alg}=4$ group only no theorem from original J_1, J_2	F32 $d_{alg}=4$ strict cone $\eta = J_2g$ semigroup branch

strict cone: $\eta = J_2g$ (4/32 per panel)
strict cone: $\eta = ij_2g$ (4/32 per panel)
first screen: cone empty by Lemma 1 (16/32)
second screen: remaining cone empty by Lemma 2 (8/32)
outlined group-only cell: time-reversal theorem proves the weight nonnegative

Primary classification: 8 strict-cone classes and 24 group-only classes per panel. Time-reversal positivity is tested only for the group-only cells: 8 bosonic and 4 fermionic classes have a verified physical representation using their original J_1, J_2 pair. Blue bosonic cells use fixed-N weights.